Smart Snowplow

Do you all have any relatives who live in snowy areas? If so, did you know that every year in the United States alone, more than 1300 people are killed every year, and over 116,800 people are injured annually due to snow plow accidents. Now imagine a world where that number could be reduced by double, triple, or maybe even tenfold!

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About Our Project

Our project is a smart, and self driving snowplow, that is designed to plow snow through fields, while also avoiding obstacles, meaning there is no driver needed.

Less Snow = Less Accidents!

The Problem We Solved

This solves a major problem, since tons of cities in the US are facing a large shortage of snowplow drivers, and our project helps solve that problem with a driverless snowplow that keeps plowing and changes direction once an obstacle is detected.

This can also improve safety by largely reducing the amount of snow plow crashes as well. This can provide major assistance to areas such as streets, or fields since it can repeatedly plow snow, with no break.

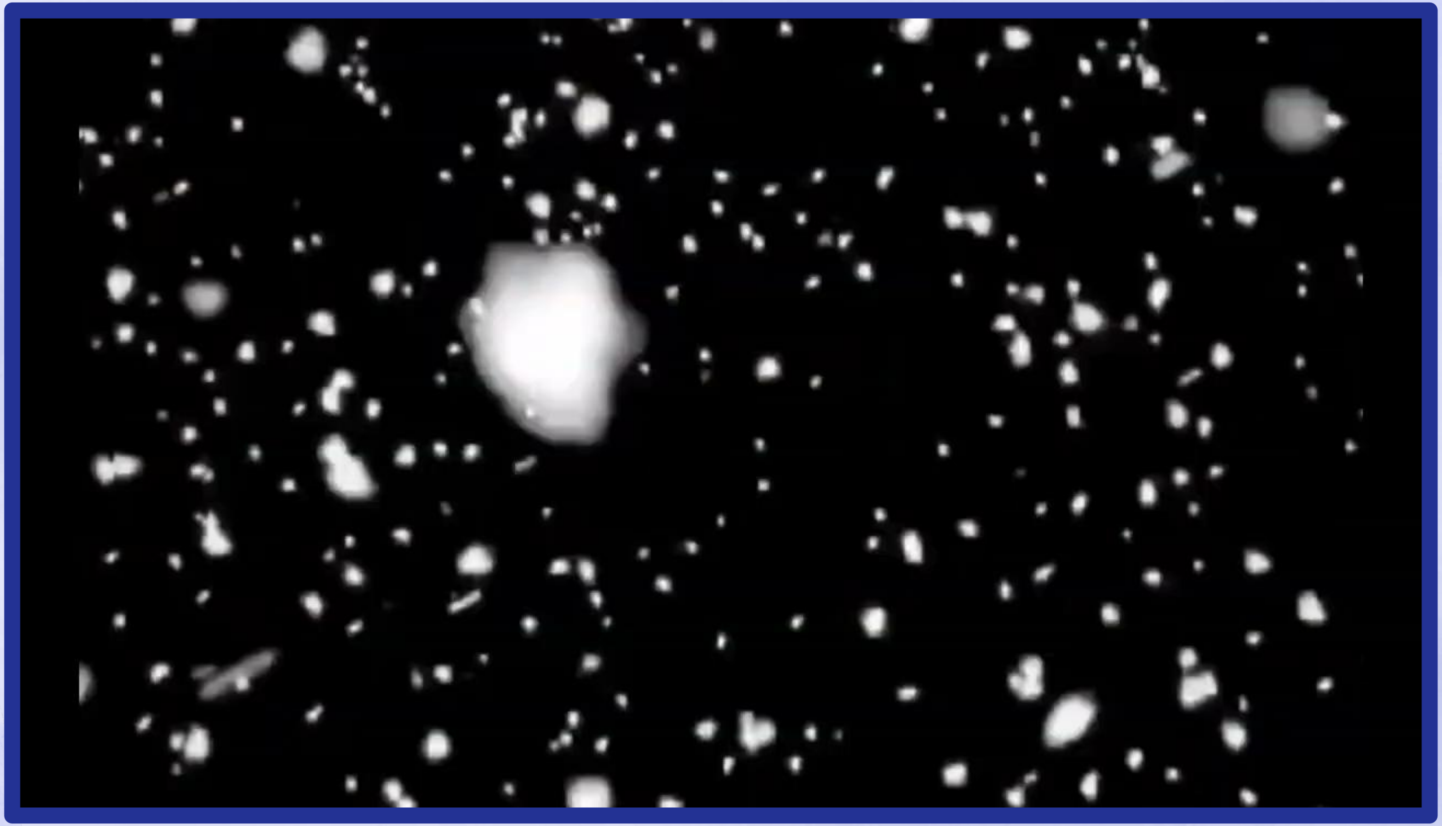
Code Showcase

This code works by using a continuous while true loop(infinite loop) to check for blue cones. If the robots vision sensor detects a blue cone it will look at how big the cone appears through the sensor using its width and height data, and based on that it will know how close the cone is, the bigger width means it closer, and smaller means its farther away. If the cone is large enough (meaning its close), the sdv stops, turns to avoid it, and then redirects itself back to its original direction before going straight.

```python
#region VEXcode Generated Robot Configuration
from vex import *
import urandom
import math

brain = Brain()

brain_inertial = Inertial()
motor_1 = Motor(Ports.PORT1, False)
motor_5 = Motor(Ports.PORT5, True)
# vex-vision-config:begin
vision_4__BLUECONE = Signature(1, -2813, -2427, -2620,7159, 7905, 7532,2.5, 0)
vision_4__REDCONE = Signature(2, 7207, 7583, 7395,-567, -255, -411,2.5, 0)
vision_4__GREENCONE = Signature(3, -5143, -4627, -4885,-2453, -1857, -2155,2.5, 0)
vision_4 = Vision(Ports.PORT4, 97, vision_4__BLUECONE, vision_4__REDCONE, vision_4__GREENCONE)


wait(100, MSEC)

def initializeRandomSeed():
    wait(100, MSEC)
    xaxis = brain_inertial.acceleration(XAXIS) * 1000
    yaxis = brain_inertial.acceleration(YAXIS) * 1000
    zaxis = brain_inertial.acceleration(ZAXIS) * 1000
    systemTime = brain.timer.system() * 100
    urandom.seed(int(xaxis + yaxis + zaxis + systemTime))

initializeRandomSeed()

from vex import *


distance = 70

def when_started1():

    motor_1.set_velocity(18, RPM)
    motor_5.set_velocity(18, RPM)


    brain_inertial.calibrate()
    while brain_inertial.is_calibrating():
        sleep(50)

    brain_inertial.set_heading(0, DEGREES)

    while True:

        objects = vision_4.take_snapshot(vision_4__BLUECONE)

        if objects and len(objects) > 0:
            cone = objects[0]

            if cone.width > distance:

                motor_1.stop()
                motor_5.stop()


                brain_inertial.set_heading(90, DEGREES)
                motor_1.spin(FORWARD)
                motor_5.spin(REVERSE)
                wait(1.5, SECONDS)


                brain_inertial.set_heading(0, DEGREES)
                motor_1.spin(FORWARD)
                motor_5.spin(REVERSE)
                wait(1.5, SECONDS)


                motor_1.spin(FORWARD)
                motor_5.spin(FORWARD)
            else:

                motor_1.spin(FORWARD)
                motor_5.spin(FORWARD)
        else:

            motor_1.spin(FORWARD)
            motor_5.spin(FORWARD)

        wait(0.2, SECONDS)


when_started1()
```

Blue cones= Obstacles

Our Project In Action



Thanks for Listening!